Website & Application Design Knowledge Base

A generic, portfolio-safe knowledge source for a Retrieval-Augmented Generation (RAG) conversational assistant. It provides design guidance and examples that can be retrieved while gathering project requirements.

1. Project Discovery

A conversational assistant should gather requirements progressively rather than asking for every detail at once. Useful categories include project type, purpose, target audience, visual style, core features, content needs, technical requirements, and optional special requests.

Project purpose

Clarify what the product should accomplish and what outcome the user expects. Examples include selling products, presenting professional work, publishing educational content, generating leads, or promoting an event.

Target audience

Identify the primary users and their needs. Consider age group, professional context, technical familiarity, location, and the actions the audience should be able to complete.

Requirement gathering principle

Ask one focused question at a time. Suggestions can be offered when useful, but suggestions should not become confirmed requirements unless the user accepts them.

2. Website Type Guidance

E-commerce

Common needs include product discovery, product details, search or filtering, shopping cart, checkout, account management, and clear purchase actions.

Portfolio

Common goals include presenting projects or work samples, communicating professional identity, and providing simple contact options. Visual hierarchy should keep attention on the work.

Blog or Content Site

Common needs include article organization, categories or search, readable typography, media presentation, and clear navigation between content items.

Business or Service Site

The interface should communicate business value clearly and make actions such as contacting the company, requesting a service, or booking an appointment easy to find.

Event Site

Typical information includes event details, schedules, speakers or performers, registration or ticketing, location, and important dates.

3. Visual Design Principles

Minimalist

Use generous whitespace, limited decorative elements, a restrained color palette, and strong visual hierarchy.

Modern

Use clear typography, balanced spacing, contemporary layouts, and consistent components without unnecessary complexity.

Vibrant

Use stronger colors and visual accents while preserving readable text, sufficient contrast, and clear hierarchy.

Professional / Corporate

Prioritize clarity, consistency, credibility, and structured information.

Artistic

Allow expressive typography, imagery, and composition while maintaining understandable navigation and content structure.

4. Layout & Usability

- Navigation should be predictable and consistent across pages.
- Consistent spacing creates visual grouping and improves readability.
- Use a clear typography hierarchy for headings, body text, labels, and actions.
- Layouts should adapt to desktop and mobile screen sizes.
- Accessibility considerations include sufficient contrast, meaningful labels, and clear interaction states.

5. Functional Features

- Search: useful when users need to locate specific content or products.
- Authentication: include accounts when they provide a clear product benefit.
- Contact forms: collect only information necessary for the intended communication.
- Gallery: useful for visually driven portfolios, products, events, or creative work.
- Booking: appropriate when users need to select an appointment, service, or time slot.
- Payments: required when the product involves online purchases or ticket transactions.
- Newsletter: useful when ongoing communication is part of the product or business goal.
- Multilingual support: define the required languages and consider how content and layout adapt.

6. Content Requirements

Content planning should identify what information must appear on the site and who provides it. Typical content may include text, images, videos, product information, project descriptions, testimonials, FAQs, schedules, or contact information.

The assistant should distinguish between content explicitly requested by the user and optional suggestions. A recommendation should not become a confirmed requirement unless the user agrees.

7. Third-Party Integrations

Potential integrations include CRM platforms, email marketing services, social media feeds, analytics tools, payment providers, calendar systems, booking services, and ticketing platforms. The assistant should ask whether an integration is required rather than assuming one.

8. Example Requirement Patterns

Online bookstore

Project type: e-commerce. Goal: sell books online. Audience: general readers. Style: minimalist. Features such as search, product categories, cart, and checkout should only be treated as confirmed when requested.

Personal portfolio

Project type: portfolio. Goal: showcase professional work. Audience: employers and clients. Style: modern. Content may include projects, skills, experience, and contact information.

Technology blog

Project type: blog. Goal: publish technology articles. Audience: students and technology professionals. Useful capabilities may include categories, search, article pages, and newsletter signup.

9. Grounding Guidelines for the Assistant

- Use retrieved knowledge as supporting context, not as a substitute for the user's stated requirements.
- Do not invent features, business goals, audience characteristics, or technical constraints.
- Ask one focused question at a time during requirement gathering.
- If retrieved information conflicts with an explicit user requirement, preserve the user's stated requirement.
- Treat optional ideas as suggestions rather than confirmed requirements.
- Keep responses concise and directly related to the current project topic.

10. Domain & Hosting Considerations

Domain and hosting are separate deployment concerns. A project may already have a domain and hosting provider, or the user may need to select them later. The assistant should collect this information when relevant rather than assuming a provider.

Portfolio note: This knowledge base is a newly created generic example for demonstrating a RAG pipeline. It is not a company document and contains no proprietary company data.